Name: Dhriti Joshi **Enrolment No:** 23FE10CAI00189	MANIPAL UNIVERSITY JAIPUR

Even Semester MTE Examination, March 2025
Faculty of Engineering, School of Computer Science and Engineering
Department Of AIML
B.Tech. (CSE) – AIML

Course Code: AIM2241	**Course: Cloud Computing**	**Semester: 4th**
Time: 1.5 hrs.		**Max. Marks: 30**

Instructions: All questions are compulsory.
Missing data, if any, may be assumed suitably.
Calculator is not allowed.

SECTION A (Memory Based Questions)

S.No		Marks	CO
Q 1	List of two real-world applications of cloud computing.	Max Marks 6 (2*3)	CO1
Q 2	Mention at least four tools used for cloud service monitoring.		CO2
Q 3	Match the following: AWS – Microsoft Azure – Salesforce GCP – Amazon Salesforce Cloud – Google		CO2

SECTION B (Concept Based Questions)

Q 4	How do HDFS and GFS handle data block size and metadata management differently, and what are the implications for scalability and fault recovery?	Max Marks 16 (4*4)	CO3
Q 5	Choose one cloud monitoring tool for a large-scale multi-cloud infrastructure (AWS, Azure, and GCP), which tool would you select and why? Explain your answer with proper justification.		CO4
Q 6	How does the underlying shuffle and sort phase in MapReduce impact performance in cloud environments with high network latency?		CO4
Q 7	In a scenario where a multinational company handles highly sensitive customer data while also requiring scalable computing power for AI-driven analytics, which deployment model (or combination) would be the most effective, and why?		CO3

SECTION-C (Analytical Based Questions)

Q 8	During peak financial events (e.g., Black Friday sales, IPO launches, or high-frequency trading), the institution dynamically wants to scale its cloud infrastructure to handle extreme transaction loads without compromising latency or security. Identify the situation with respect to these three models: a) IaaS b) PaaS c) SaaS	Max Marks 8	CO2 CO3

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Faculty of Engineering, School of Computer Science and Engineering
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Course Code: AIM2241	**Course: Cloud Computing**	**Semester: IV**
Time: 3 hrs.		**Max. Marks: 80**

Instructions: All questions are compulsory.
Missing data, if any, may be assumed suitably.
Calculator is allowed.

SECTION A (Memory Based Questions)

S. No.		Marks	CO
Q A1	What are the security challenges associated with Cloud Computing?	Max Marks 5*2=10	CO1
Q A2	Distinguish between Parallel Computing, Distributed Computing		CO1
Q A3	Discuss in brief about the cloud computing and data security.		CO3
Q A4	Mention the applications of high performance and high throughput systems.		CO4
Q A5	Draw the architecture of Windows Azure.		CO2

SECTION B (Concept Based Questions)

Q B1	Discuss the concept of virtualization in detail. Explain its significance and primary benefits in modern computing environments. Additionally, enumerate the various types of virtualization and elaborate on how each type contributes to efficient utilization of computing resources.	Max Marks 5*6=30	CO3
Q B2	Cloud computing has significantly transformed the landscape of scientific research and computational modeling. Discuss the fundamental advantages that cloud computing technology offers to scientific applications. In your response, highlight the specific features of cloud infrastructure that make it particularly suitable for research, large-scale data analysis, and scientific simulations. Support your answer with relevant examples.		CO4
Q B3	Discuss the fundamental components of the Infrastructure-as-a-Service (IaaS) model in cloud computing. How do these components contribute to scalability, flexibility, and efficient resource management? Support your explanation with detailed examples of real-world IaaS implementations, and analyze how these examples demonstrate the practical capabilities and advantages of the IaaS model.		CO1

Q B4	You are tasked with designing a scalable web application for a startup that expects rapid growth in user traffic. The company has chosen to deploy the application using Google App Engine. Explain the architecture of Google App Engine and discuss how its components help support scalability, reliability, and efficient resource management.		CO2
Q B5	During a seminar on cloud computing, two terms were introduced that left participants seeking clarification: cloud scientific applications and third-party cloud services. Define and explain the following terms with relevant examples: (a) Cloud scientific application (b) Third-party cloud services		CO4
	SECTION-C (Analytical Based Questions)		
Q C1	Cloud computing has emerged as a transformative technology across various sectors, including business, healthcare, education, and scientific research.Provide a comprehensive overview of the diverse applications of cloud computing in different industries. How has cloud computing revolutionized these sectors in terms of efficiency, scalability, and innovation? Additionally, discuss the specific advantages that cloud computing offers to scientific applications. Why is cloud infrastructure particularly well-suited for handling research workloads, simulations, and large-scale data analysis in the scientific domain?	Max Marks 20	CO2
Q C2	A retail company wants to analyze customer purchase data across its online and offline platforms to improve its sales strategy. Due to large data volume and variety, the company is exploring cloud-based solutions for data analysis. Explain how cloud computing supports data analysis applications and discuss the benefits it offers in this scenario.		CO5
	SECTION-D (Case Study/Application based Question)		
Q D1	A multinational company is planning to migrate its IT infrastructure to cloud. The management team is evaluating different cloud deployment models to find the most suitable option based on factors such as data sensitivity, user access, and administrative control. At the same time, they are concerned about potential challenges such as data security, service reliability, and vendor lock-in. Classify and describe the various types of cloud deployment models (public, private, hybrid, and community). Compare them in terms of access, management, and security. Additionally, identify and elaborate on some of the key challenges the company may face during and after cloud migration. Suggest appropriate strategies to overcome these challenges and ensure a smooth transition.	Max Marks 20	CO3
Q D2	A medium-sized manufacturing company is struggling with disconnected data between its sales, inventory, and customer service departments. They are considering implementing CRM and ERP systems to streamline their operations. Explain the concepts of Customer Relationship Management (CRM) and Enterprise Resource Planning (ERP) and how integrating these systems could solve the company's problems.		CO5